

SAHIL LENKA

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EDUCATION

Manipal University Jaipur , Jaipur, Rajasthan	Expected 2028
<i>B.Tech in Computer Science and Engineering (Artificial Intelligence and Machine Learning)</i>	
Don Bosco School , Bandel	Class XII 2024, Class X 2022

EXPERIENCE

Web Development Intern , IOTA Studio AI (AIC MUJ incubated startup), Remote	Apr 2025 - Aug 2025
- Built and deployed the company's official full-stack website using React.js, Node.js and REST APIs, with 10+ production commits merged into the main codebase. - Designed the REST API with a clean frontend and backend separation, using MongoDB and SQL for content.	
Software Development Intern , Sundarone Private Limited (sundaronehostel.in)	Feb 2024 - May 2024
- Integrated the Razorpay payment gateway into the booking flow, contributing to a 15% increase in bookings. - Led UX iteration cycles that increased user retention by 20%.	

PROJECTS

EvidenceRAG Python, FastAPI, React, TypeScript, Ollama, Sentence Transformers	github.com/Sahil-u07/evidencerag
- Built a local-first document question-answering system (RAG) that answers only from retrieved evidence and abstains when the evidence is insufficient. - Implemented hybrid dense and BM25 retrieval with Reciprocal Rank Fusion and cross-encoder reranking, reaching an MRR of 1.000 on a 20-question benchmark. - Added NLI-based verification of every cited claim; the backend is covered by 97 automated tests.	
RoadGuard AI (in progress, team of 3) Python, YOLOv8, Streamlit, OpenCV	github.com/Sahil-u07/roadguard-ai
- Detects potholes, cracks and speed bumps in road images and video with a fine-tuned YOLOv8 object detection model. - Built a tested dataset-merge pipeline that maps differently labelled YOLO and Pascal-VOC datasets onto one class list. - Pilot crack model reached mAP@0.5 of 0.606 and F1 of 0.651; full training on the merged dataset is in progress.	
ProjectMUJToppers React.js, Node.js, MongoDB	mujtoppers.in
Team-built full-stack platform showcasing university toppers, with authentication, an admin dashboard and a student database; developed with Git feature branches and live in production.	

OPEN SOURCE CONTRIBUTIONS (33 MERGED PULL REQUESTS)

Beehive , KathiraveluLab Python, Flask, React, GitHub Actions	13 merged PRs
- Built a static security scanner that maps API auth decorators and flags unprotected endpoints by file and line, identifying 6 vulnerabilities (1 Critical, 4 High, 1 Medium). - Hardened secret-key and environment config validation, disabled debug mode in production, and added React component tests (Vitest) and linting to CI.	
Concore , ControlCore-Project Python, C++, MATLAB, ZeroMQ	13 merged PRs
- Built the concore command-line tool (init, run, validate, status, stop) and a workflow inspect command with JSON export; added unit tests and GitHub Actions CI. - Fixed a shared-memory race between concurrent writers using a seqlock and a POSIX semaphore, and fixed simulation-time bugs in the C++ and MATLAB bindings.	
Diomede , KathiraveluLab Python, pynetdicom, Docker Compose	5 merged PRs
Implemented dynamic DICOM routing across multiple Orthanc nodes with weighted destination scoring and a health-checker daemon; added routing regression tests.	
GNU Radio Python, Qt, C++	2 merged PRs
Fixed redundant move actions and signals in the GNU Radio Companion Qt editor; updated WAV file block documentation and Python bindings for libsndfile support.	

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C++, Java, SQL, HTML/CSS
Frameworks: React.js, Node.js, Express.js, Flask, FastAPI, Streamlit, TailwindCSS
AI/ML: Retrieval-Augmented Generation (RAG), YOLOv8, Sentence Transformers, Ollama
Databases and Tools: MongoDB, PostgreSQL, Git, GitHub Actions (CI/CD), Docker (basics), Linux, Postman, Vercel, Netlify
Other: REST APIs, JWT Authentication, RBAC, Security Hardening, Full-Stack Architecture

ACHIEVEMENTS

- HackRx 6.0 (Bajaj Finserv):** Top 35 out of 10,000+ teams in a national-level hackathon.
- Smart Delhi Ideathon:** 3rd place in category at a government-backed tech innovation competition.
- 3x Hackathon Finalist:** reached the finals of three university-level and national hackathons.